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Research on security strategies in the power wireless private network

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Abstract

This paper presents a secure communication access authentication technology for power wireless private network. At the physical layer, the device is identified based on physical radio frequency fingerprint authentication, which improves the security of device access. At the same time, the idea of distributed authentication is used to reduce the device authentication pressure of the gateway on the core network side and improve the speed of access device authentication.

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1. Introduction

With the wide application of wireless private network in power system, it provides convenient access means for power distribution automation, source network load storage interaction and other control services, and power information acquisition, mobile operation, video monitoring and other management services [1-2]. Power wireless private network (PWPN) inherits the mature wireless networking, construction is convenient, flexible application of advantage, at the same time, its frequency, the equipment, network dedicated to avoid the wireless public network bandwidth, delay, the limitation of business interruption rate, safety, reliability, can effectively supplement the cable transmission network and efficient to solve electric power communication "the last kilometer" problem access, through electric power communication network "nerve endings," a traditional wired communications and wireless public network communication incomparable advantages [3-5].

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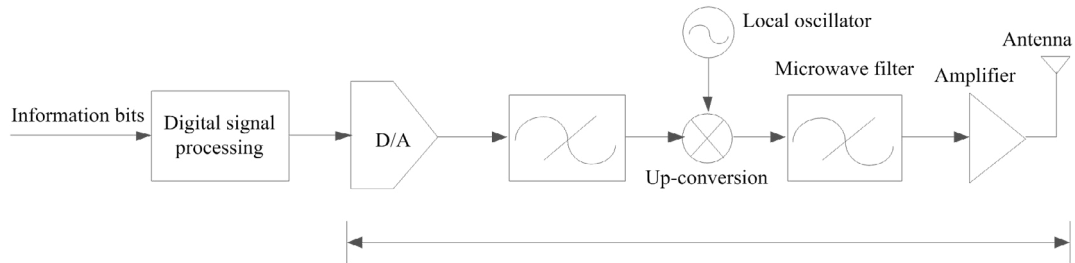


Fig. 1. The procedure of transmit information processing

However, with the rapid increase of wireless private network access terminals, the source network store, distribution change monitoring, mobile applications and other kinds of mixed business unified access scenario, the current widely used by way of centralized processing core network will not be able to meet the real-time requirement, the load control class electric power business power wireless private network also inherited the channel of the wireless system and open network sharing, terminal mobile features, also proposed the challenge to the safety of the electric power business [6-7].

The centralized authentication method cannot meet the real-time requirements of the source network load storage and other services. In power wireless private network communication mode, centralized authentication by calling the master station core network computing resources are needed to realize bidirectional authentication and wireless private network terminal, large scale power terminal access authentication will take up a lot of network resources, and results in large computation overhead and storage overhead, can meet the real-time load control class business requirements.

The communication full path security monitoring system for power wireless private network is missing. Power wireless private network has not yet been built for the terminal - base station - core network - the main "architecture of an end-to-end security communication monitoring system and the terminal access layer, system layer and the business management the lack of effective monitoring analysis system, be able to power wireless private network communications, an abnormal business for rapid positioning, found the problem accurately.

2. Radio frequency fingerprint

2.1 Generation of RF fingerprint

Everyone is born with a unique biometric fingerprint, and even twins have their own. It can be used as an important basis for identification. Similar to biological fingerprint, wireless devices also have their unique fingerprints. The difference mainly comes from the hardware device manufacturing procedure which occurs inevitably. So, the fingerprint characteristics can be as a key indicator of device identification authentication. This technology is based on the physical mechanism of certification which is not easy to imitate [8-9].

According to the classic structure of the signal transmitting system as shown in Fig. 1, the generation of RF fingerprint will be discussed in detail in the following. In general, wireless transmit system includes two parts, the baseband signal is generated by digital signal processing first. Then the digital signal will be transmitted through the analog circuit with analog modulation. The hardware characteristics of the wireless transmit system will be included in the transmitted RF signal. Finally, the signal is received by the receiving device so that the source of RF fingerprint difference comes from the difference of analog circuit.

Device tolerance refers to the irretrievable tolerance between the actual electrical parameters of an electronic device and the standard parameters. Because in the layout and the use of the corresponding technology to combine the electronic components and related materials in the manufacturing process, the processing technology cannot be

accurate to ensure that each produced hardware is exactly the same. The device tolerance mainly includes manufacturing tolerance and drift tolerance.

2.2 Characteristics of RF fingerprint

Extracting a feature or set of features from a recognizable signal that can be used to identify a wireless device. In order to ensure the realizability and good identification performance of RF fingerprint in practical application, the RF fingerprint, as the identity identification mark of wireless communication equipment, must meet the following five characteristics.

1) RF fingerprint must be universal. The precondition for the wide application of RF fingerprint is universality, which includes the universality of transmitting device and receiving device. Firstly, the rf fingerprint that meets the requirements can be extracted from the transmitting signals of the same type of device, which is called the universality of transmitting device. Secondly, the use of different receiving devices will not affect the wireless signals collected by the fingerprint system. And it only contains features related to the transmitting devices to be identified, which is called the universality of receiving devices.

2) RF fingerprints must be unique. We know that even identical, difficult-to-distinguish twins' biometric fingerprints are unique. Just like the same wireless device produced by the same manufacturer using the same production device, although they are very similar in terms of hardware, they will not be exactly the same. Therefore, only the unique RF signal extracted from the RF signal can be used for the identification of wireless device.

3) RF fingerprints must be robust. In the actual application environment, temperature, humidity and so on will have a relatively large difference. And even strong noise interference, and multipath effect will also affect the performance of device identification. The robustness of RF fingerprint means that in different communication environments, RF fingerprint can resist the influence of the above factors. In a word, no matter how the communication environment is changed, RF fingerprint will not fail or change.

4) RF fingerprint must be independent. The wireless signal sent by the transmit device carries different modulation information, while the independence means that the rf fingerprint has nothing to do with the transmitted data. The data transmitted by each signal may not be the same, but the fingerprint features extracted from it should be the same to reflect the hardware differences of the device.

5) RF fingerprint must meet short-term invariability. Due to the influence of temperature, humidity, dust and other factors in the operating environment, the circuit boards of wireless devices will be aging or damaged over time. In the case of the above changes in the transmit device hardware, the real-time extracted RF fingerprints will also change accordingly, resulting in a mismatch between the fingerprints registered in the RF fingerprint database. Fortunately, the aging process of general wireless devices is relatively long, and the range of hardware parameters floating caused by aging is relatively small, so RF fingerprint is relatively stable.

3. Architecture of distributed security gateway based on RF fingerprint

3.1 Traditional security strategies

The new generation security architecture of 3GPP presents the construction of a new group AKA protocol based on reversible Hash tree. The scheme is based on tree structure master key storage devices, multiple IoT devices, can be a one-time certification and can dynamically on the front end calculation of certification and the back-end server traffic between adjustment, can be directly deployed to the existing communication system, and through formal tools validation, can provide the bidirectional authentication and key device to the network privacy confidentiality, device such as safety properties.

In addition, in order to improve access efficiency, the recommended encryption scheme given by 3GPP in TR33.899 is DHIES and its variant ECIES on ECC [10-11]. 3GPP also provides identity-based encryption (IBE) and property-based encryption (ABE) solutions that directly encrypt the user's identity or encrypt the identity with a private key and global public key bound to a public property.

For the access of illegal terminal, the authentication process of the device is blocked through the function of authentication gateway. When the distributed safety net is closed, two mechanisms are used to protect the identity of access terminals: temporary identity identification and encrypted permanent identity identification. Among them, temporary identity identification mainly refers to the interface in the air, try to use a constantly updated and temporarily assigned identity identification, in order to replace the permanent identity identification, in order to greatly reduce the probability of permanent identity identification intercepted by the air.

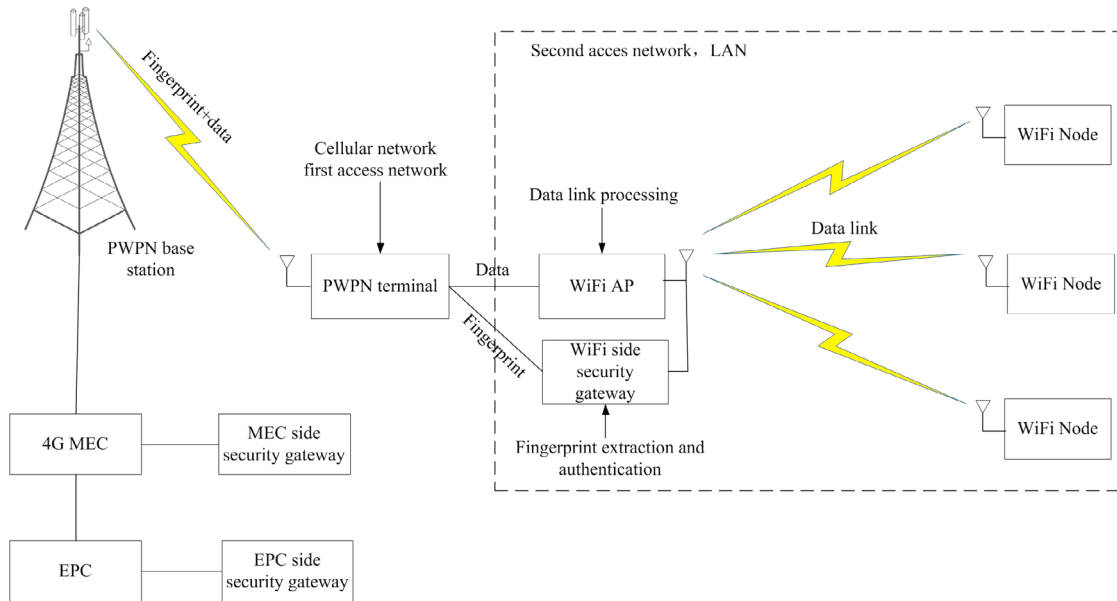


Fig. 2. Architecture of distributed security gateway in PWPN

The encryption of permanent identity is referred to in the air interface, as far as possible to encrypt the identity of the transmission. In the distributed authentication gateway, the illegal identity can be blocked directly whether it is temporary or permanently encrypted.

3.2 Authentication based on physical fingerprint

Currently, the fingerprint of wireless devices can be divided into two types: turn-on wireless device fingerprint based on transient signal and steady-state wireless device fingerprint based on steady state signal. Wireless device fingerprint identification technology can be divided into wireless device fingerprint identification technology based on transient signal, wireless device fingerprint identification technology based on steady state signal and wireless device fingerprint identification technology based on spectrum characteristics.

In the case of two-stage networking in force system, the architecture of WiFi security gateway system based on physical fingerprint required by PWPN is shown in Fig. 2. The system is a distributed deployment, contains the EPC side gateway, MEC side gateway and WiFi local side gateway, these gateways use RF fingerprint as the authentication basis. It also can realize the blocking of an device operation. As shown in Fig. 3, The detail steps are as follows:

Step 1: The received signal is detected by a correlator, and the peak of correlator is used to determine whether the IEEE 802.11a/g signal is received. The physical fingerprint of the device is obtained through the training sequence. This is done on a separate listening device, not on a traditional WiFi access point. So, the stand-alone device can

listen to the physical layer signals of devices within multiple WiFi access points. If no correlator peak is detected, it is ready to enter the next cycle of correlator peak detection.

Step 2. Obtain the MAC address of the device by decoding the received data. The MAC address is decoded by an independent listening device using the received information, rather than the MAC information of the WiFi access point.

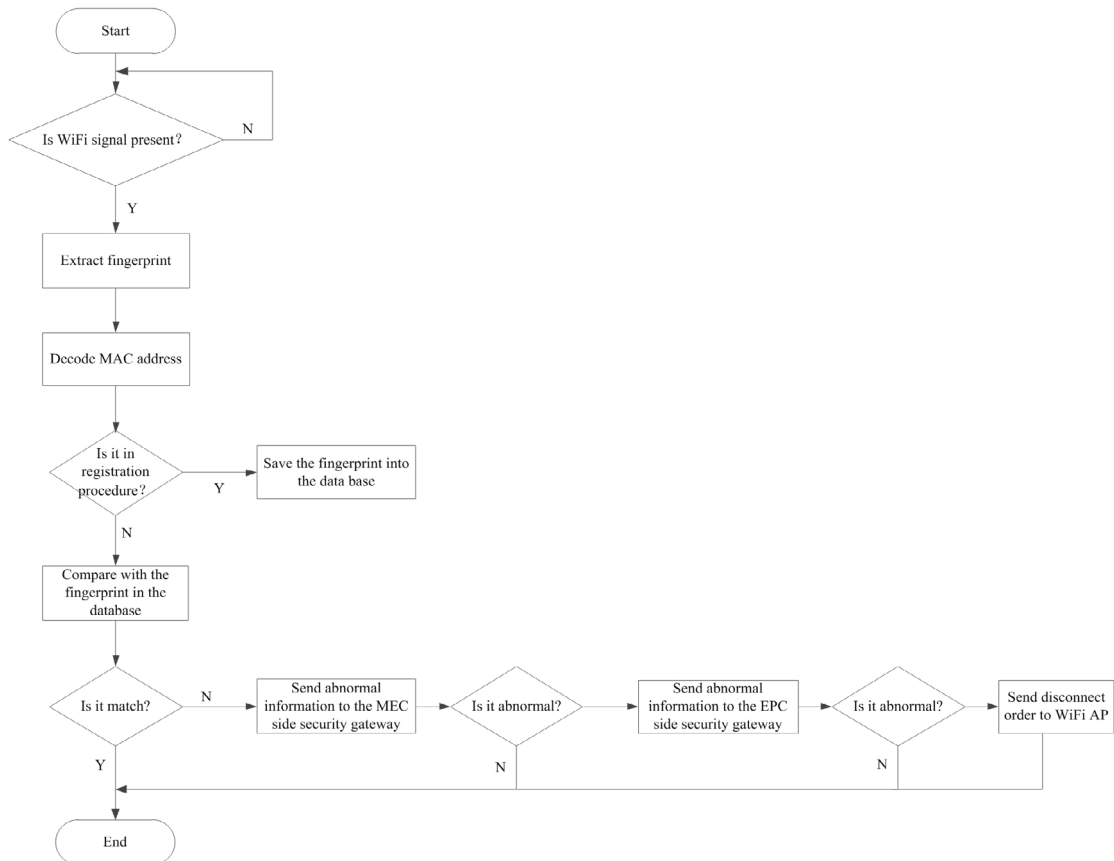


Fig. 3. Illustration of access procedure based on the physical fingerprint authentication

Step 3. The MAC address obtained in Step 2 is used to determine the corresponding fingerprint in the local side fingerprint database and compare with the detected fingerprint. If the similarity is lower than the decision threshold, the access device is considered abnormal. If the similarity exceeds the decision threshold, no further operation will be done, and the device will be deemed to have no abnormal access this time.

Step 4. Store and comprehensively determine the abnormal information transmitted from the WiFi edge. If a device continuously produces more than N ($N > 1$) abnormal fingerprint samples in a given time period, the device will be identified as an abnormal device. And then the stored abnormal fingerprint samples and their MAC addresses will be sent to the core network side gateway. If no more than N abnormal fingerprint samples occur in a given period of time, the device is considered as a normal device. The variable N here can be determined by practice.

Step 5. After the core network gateway receives the abnormal information reported by MEC side gateway, it makes a comprehensive judgment based on the situation of the whole network, such as MAC address change and

device replacement record. If the decision is abnormal, it will send device blocking instruction to MEC side gateway. If the device is an access device, the change fingerprint instruction will be sent to the MEC side gateway.

Step 6, At the MEC side gateway, according to the received instructions, it will find the abnormal device's WiFi access point according to the MAC address, and then sent blocking instructions. The blocking instruction is valid only for a limited time, such as continuous block 10 minutes. After 10 minutes, the device can be allowed to access. It is used to prevent the device permanently unable to access caused by error block instruction.

4. Conclusions

This paper review of the demand for electric power wireless private network, and the distributed power wireless private network terminal access authentication first. Then we proposed a distribute authentication solution to reduce the authentication delay and the authentication pressure of EPC. The access authentication based on the fingerprint can improve the level of power wireless private network safety, ensure the safety of electric power communication end-to-end transmission and the reliable operation of the system.

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